2.2: Operational Amplifiers (Op-Amps)

1. Introduction to Operational Amplifiers

Operational Amplifiers, commonly known as Op-Amps, are versatile integrated circuits that are fundamental building blocks in analog electronics. They are essentially high-gain, differential voltage amplifiers. Originally developed for performing mathematical operations in analog computers, their applications have expanded significantly to include filtering, signal conditioning, and various control systems.

1.1 Key Characteristics of an Ideal Op-Amp

An **ideal op-amp** is a theoretical model that simplifies the analysis of op-amp circuits. While real op-amps deviate from this ideal, understanding the ideal characteristics provides a strong foundation for circuit design and analysis. The key characteristics are:

- **Infinite Open-Loop Gain (A_{OL}):** This means even a tiny difference in voltage between the input terminals will result in a very large output voltage. In practical terms, this implies that the input differential voltage ($V_d = V_+ - V_-$) is driven to zero when negative feedback is applied.
- **Infinite Input Impedance (Z_{in}):** No current flows into the input terminals of the op-amp ($I_+ = I_- = 0$). This simplifies circuit analysis as the op-amp inputs do not load the preceding stage.
- **Zero Output Impedance (Z_{out}):** The op-amp can supply any amount of current to the load without any voltage drop across its internal resistance. This means the output voltage is independent of the load.
- **Infinite Bandwidth:** The op-amp can amplify signals of any frequency without attenuation.
- **Zero Offset Voltage:** When both input terminals are at the same voltage, the output voltage is exactly zero.

These ideal characteristics lead to two fundamental rules for analyzing op-amp circuits with negative feedback:

1. **No current flows into the input terminals:** $I_+ = I_- = 0$
2. **The voltage difference between the input terminals is zero:** $V_+ = V_-$ (Virtual Short Circuit)

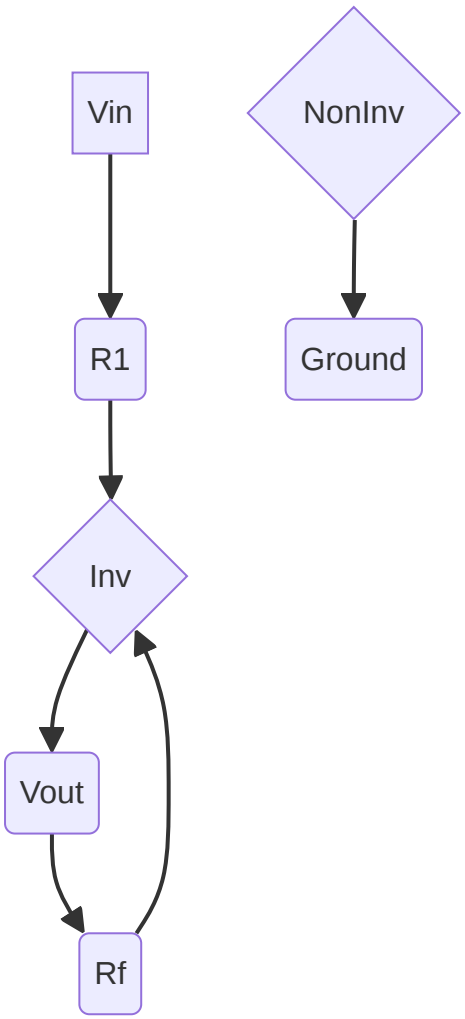
2. Common Op-Amp Configurations

Op-amps are rarely used in their open-loop configuration due to their extremely high gain, which makes them unstable and prone to saturation. Instead, they are typically used with **negative feedback**, where a portion of the output signal is fed back to the inverting input. This feedback stabilizes the amplifier and allows for precise control over its gain and other characteristics.

2.1 Inverting Amplifier

The inverting amplifier configuration is one of the most fundamental op-amp circuits. In this configuration, the input signal is applied to the inverting (-) input terminal through a resistor, while the non-inverting (+) input terminal is grounded. The output is out of phase with the input.

Circuit Diagram:



Analysis (Ideal Op-Amp Assumptions):

1. **Virtual Short:** $V_- = V_+ = 0V$ (since V_+ is grounded).
2. **No Current into Inputs:** $I_{R1} = I_{Rf}$

   Current through R_1: $I_{R1} = \frac{V_{in} - V_-}{R_1} = \frac{V_{in} - 0}{R_1} = \frac{V_{in}}{R_1}$

   Current through R_f: $I_{Rf} = \frac{V_- - V_{out}}{R_f} = \frac{0 - V_{out}}{R_f} = -\frac{V_{out}}{R_f}$

   Equating the currents: $\frac{V_{in}}{R_1} = -\frac{V_{out}}{R_f}$

   Voltage Gain (A_v): $A_v = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$

Key Characteristics:

- The output voltage is inverted (180 degrees out of phase) with respect to the input.

- The gain is determined by the ratio of the feedback resistor (R_f) to the input resistor (R_1).
- The input impedance of the circuit is approximately R_1.

Example:

Design an inverting amplifier with a gain of -10. If $R_1 = 1k\Omega$, what should R_f be?

Solution:

$A_v = -\frac{R_f}{R_1}$

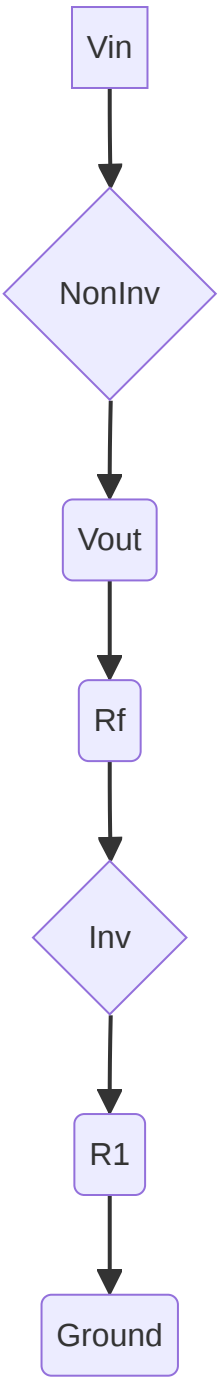
$-10 = -\frac{R_f}{1k\Omega}$

$R_f = 10 \times 1k\Omega = 10k\Omega$

2.2 Non-Inverting Amplifier

In the non-inverting amplifier configuration, the input signal is applied directly to the non-inverting (+) input terminal, and the feedback network is connected from the output to the inverting (-) input terminal. The output is in phase with the input.

Circuit Diagram:



Analysis (Ideal Op-Amp Assumptions):

1. **Virtual Short:** $V_- = V_+ = V_{in}$
2. **No Current into Inputs:** $I_{R1} = I_{Rf}$

   Current through R_1: $I_{R1} = \frac{V_- - 0}{R_1} = \frac{V_{in}}{R_1}$

   Current through R_f: $I_{Rf} = \frac{V_{out} - V_-}{R_f} = \frac{V_{out} - V_{in}}{R_f}$

   Equating the currents: $\frac{V_{in}}{R_1} = \frac{V_{out} - V_{in}}{R_f}$

$V_{in}R_f = R_1(V_{out} - V_{in})$

$V_{in}R_f = R_1V_{out} - R_1V_{in}$

$V_{in}R_f + R_1V_{in} = R_1V_{out}$

$V_{in}(R_f + R_1) = R_1V_{out}$

Voltage Gain (A_v): $A_v = \frac{V_{out}}{V_{in}} = \frac{R_f+R_1}{R_1} = 1 + \frac{R_f}{R_1}$

Key Characteristics:

- The output voltage is in phase with the input.
- The gain is always greater than or equal to 1.
- The input impedance of the circuit is very high (ideally infinite), as the input signal is applied directly to the non-inverting input of the op-amp.

Example:

Calculate the output voltage of a non-inverting amplifier if $V_{in} = 0.5V$, $R_1 = 2k\Omega$, and $R_f = 18k\Omega$.

Solution:

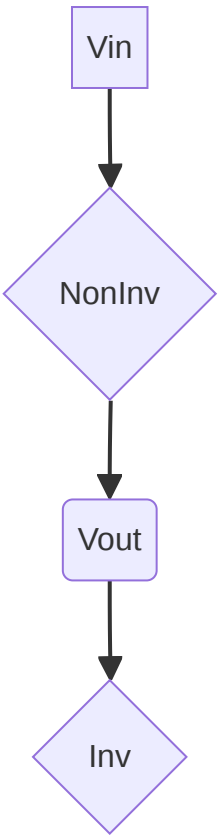
$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{18k\Omega}{2k\Omega} = 1 + 9 = 10$

$V_{out} = A_v \times V_{in} = 10 \times 0.5V = 5V$

2.3 Voltage Follower (Buffer)

The voltage follower, also known as a buffer amplifier, is a special case of the non-inverting amplifier where the entire output voltage is fed back directly to the inverting input. This results in a gain of exactly 1.

Circuit Diagram:



Analysis (Ideal Op-Amp Assumptions):

1. **Virtual Short:** $V_- = V_+ = V_{in}$
2. **Feedback:** $V_{out} = V_-$ (since the output is directly connected to the inverting input)

   Therefore, $V_{out} = V_{in}$

Key Characteristics:

- Gain is 1 ($A_v = 1$).
- Provides high input impedance and low output impedance, making it ideal for **buffering** or **impedance matching** between stages.
- It does not amplify the voltage but can amplify the current, allowing a weak source to drive a heavy load.

Example:

A sensor has a high output impedance and produces a voltage of 2V. If this sensor is connected directly to a low-impedance load, the voltage drops significantly. How can a voltage follower help?

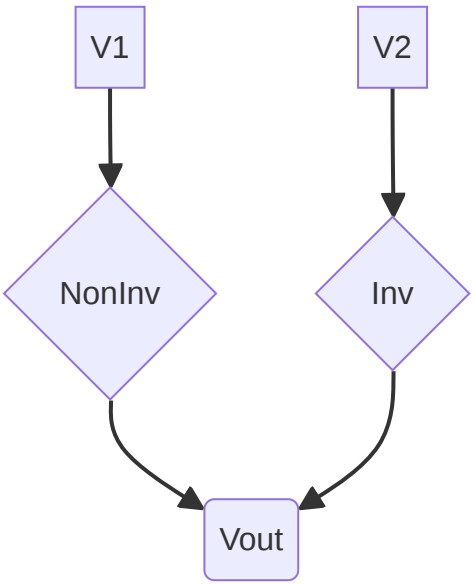
Solution:

By placing a voltage follower between the sensor and the load, the high input impedance of the op-amp will not load the sensor, so the sensor's 2V output is maintained at the op-amp's input. The low output impedance of the op-amp can then drive the low-impedance load without significant voltage drop, effectively transferring the 2V from the sensor to the load.

2.4 Comparator

Unlike the previous configurations that use negative feedback, a comparator typically operates in an **open-loop** configuration or with positive feedback. Its primary function is to compare two input voltages and produce an output that indicates which input is greater.

Circuit Diagram:



Analysis (Ideal Op-Amp Assumptions):

Since there is no negative feedback, the op-amp operates with its very high open-loop gain. The output will quickly saturate to either the positive saturation voltage ($+V_{sat}$) or the negative saturation voltage ($-V_{sat}$), which are typically close to the positive and negative power supply rails.

- If $V_+ > V_-$, then $V_{out} = +V_{sat}$
- If $V_+ < V_-$, then $V_{out} = -V_{sat}$

Key Characteristics:

- No feedback (or positive feedback for hysteresis).
- Output is a digital signal (high or low), indicating the comparison result.
- Used in applications like zero-crossing detectors, level detectors, and analog-to-digital converters.

Example:

An op-amp is configured as a comparator with $+V_{sat} = +12V$ and $-V_{sat} = -12V$. If V_1 is connected to the non-inverting input and V_2 to the inverting input, what is V_{out} when:

a) $V_1 = 3V, V_2 = 2V$

b) $V_1 = 1V, V_2 = 5V$

Solution:

a) Since $V_1(3V) > V_2(2V)$, $V_{out} = +12V$.

b) Since $V_1(1V) < V_2(5V)$, $V_{out} = -12V$.

3. Comparison of Op-Amp Configurations

Feature	Inverting Amplifier	Non-Inverting Amplifier	Voltage Follower	Comparator
Feedback Type	Negative	Negative	Negative	None (or Positive)
Input Applied To	Inverting (-)	Non-Inverting (+)	Non-Inverting (+)	Both Inputs
Gain (A_v)	$-R_f/R_1$	$1 + R_f/R_1$	1	Open-loop (very high)
Output Phase	Inverted	In-phase	In-phase	Digital (High/Low)
Input Impedance	Moderate (R_1)	Very High (Ideal: ∞)	Very High (Ideal: ∞)	High
Output Impedance	Very Low (Ideal: 0)	Very Low (Ideal: 0)	Very Low (Ideal: 0)	Very Low (Ideal: 0)
Primary Use	Signal Inversion & Gain	Signal Amplification	Buffering, Impedance Matching	Voltage Comparison

4. Practice Problems

1. An inverting amplifier has $R_1 = 2.2k\Omega$ and $R_f = 33k\Omega$. If $V_{in} = 0.1V$, what is V_{out} ?
2. Design a non-inverting amplifier with a gain of 5. If $R_1 = 10k\Omega$, what is R_f?
3. Explain the primary purpose of a voltage follower.
4. An op-amp is used as a comparator with reference voltage $V_{ref} = 0V$ at the inverting input. If a signal V_{sig} is applied to the non-inverting input, describe the output when $V_{sig} > 0V$ and $V_{sig} < 0V$.

5. Solutions to Practice Problems

1. **Solution:** $A_v = -\frac{R_f}{R_1} = -\frac{33k\Omega}{2.2k\Omega} = -15\, V_{out} = A_v \times V_{in} = -15 \times 0.1V = -1.5V$
2. **Solution:** $A_v = 1 + \frac{R_f}{R_1}\, 5 = 1 + \frac{R_f}{10k\Omega}\, 4 = \frac{R_f}{10k\Omega}\, R_f = 4 \times 10k\Omega = 40k\Omega$
3. **Solution:** The primary purpose of a voltage follower (buffer) is to provide **impedance matching** between a source and a load. It has a very high input impedance, which prevents loading of the source, and a very low output impedance, which allows it to drive a low-impedance load without significant voltage drop or signal attenuation. It provides a voltage gain of 1.
4. **Solution:** This is a zero-crossing detector. The non-inverting input is V_{sig} and the inverting input is $0V$.
   - If $V_{sig} > 0V$, then $V_+ > V_-$, so $V_{out} = +V_{sat}$.
   - If $V_{sig} < 0V$, then $V_+ < V_-$, so $V_{out} = -V_{sat}$.

References

[1] Sedra, A. S., & Smith, K. C. (2015). *Microelectronic Circuits* (7th ed.). Oxford University Press. [2] Boylestad, R. L., & Nashelsky, L. (2013). *Electronic Devices and Circuit Theory* (11th ed.). Pearson Education.